



TICLOPIDINE TABLET

Description: White, round, biconvex, non-scored film-coated tablet.

Content:

Each tablet contains ticlopidine hydrochloride 250mg

Indications:

For the maintenance of patency coronary artery by-pass grafts, maintenance of patency of access sites in patients on chronic hemodialysis, and reducing risk of thrombotic stroke (fatal or non-fatal) in patients who have experienced stroke precursors and in patients who have had a completed stroke. Because ticlopidine is associated with a risk of neutropenia/agranulocytosis, which may be life-threatening, ticlopidine should be reserved for patients who are intolerant to aspirin therapy where indicated to prevent stroke.

Pharmacology:

Ticlopidine, an inhibitor of platelet aggregation, which inhibits adenosine diphosphate (ADP)-induced binding of fibrinogen to the platelet membrane at a specific receptor site (the glycoprotein IIb/IIIa complex). Release of platelet granule constituents, platelet-platelet interactions, and platelet adhesion to the endothelium and to atheromatous plaque are inhibited. Its activity is dose-dependent. Upon discontinuation of treatment, bleeding time and other platelet function tests will return to normal within a week. Ticlopidine-induced inhibition of platelet aggregation persists for the life of the platelet. It prolongs the template bleeding time, but has no effect in usual assays of coagulation or fibrinolysis. Ticlopidine also reduces fibrinogen concentrations and blood viscosity, and increases the filterability rates of both whole blood and red cells, which may contribute to the beneficial effects in patients with vascular disease.

Ticlopidine hydrochloride is rapidly and almost completely absorbed from the gastrointestinal tract. 80% or more of a dose is absorbed. Absorption is increased when the medication is taken after a meal (20% increase of area under curve (AUC) of ticlopidine). The terminal half-life during chronic dosing is reported to be about 30-50 hours. Ticlopidine is rapidly and extensively metabolised in the liver. It is reversibly protein bound (98%), primarily to serum albumin, lipoproteins, and also to alpha-1 acid glycoprotein (15% or less). Protein binding of metabolites is about 40-50%. Approximately 60% of the administered dose is excreted in the urine and about 23% in the feces. Approximately 1/3 of the dose excreted in the feces is intact ticlopidine hydrochloride, possibly excreted in the bile.

Adverse effects:

Gastrointestinal disturbances and skin rashes are the most commonly reported side effects. Blood dyscrasias, aplastic anemia and haemorrhagic disorders have also occurred. Hepatitis, cholestatic jaundice and increase of alkaline phosphatases, transaminases & bilirubin have also been reported. Blood lipid concentration may increase during long term therapy. Severe neutropenia or agranulocytosis may occur in about 1% of patients. Thrombotic thrombocytopenic purpura, immunological reactions e.g vasculitis, lupus syndrome have also been reported.

Precautions:

Ticlopidine should be used with caution in patients likely to bleed and should not be given concomitantly with heparins, oral anticoagulants and antiplatelet drugs, as it may cause haemorrhagic complications. For patients undergoing elective surgery, treatment should wherever possible be stopped at least 10-14 days before surgery. However, in the event of emergency surgery, three means may be used alone or in combination to limit the risk of haemorrhage and prolonged bleeding time, administration of 0.5-1mg/kg of methylprednisolone IV, to be renewed; desmopressin at a dosage of 0.2-0.4ug/kg; platelet transfusions. CBC's with differential and platelet counts should be performed at onset of therapy, and regular haematological monitoring be repeated every 2 weeks for the first 12 weeks of treatment and within 15 days after discontinuation of therapy. Clinical monitoring should be done and signs & symptoms of adverse drug reactions should be observed during the first 3 months of therapy. Sign and symptoms possibly related to neutropenia (fever, sore throat, ulcerations in oral cavity), jaundice (including dark urine, light coloured stool) and thrombocytopenia & abnormal haemostasis (prolonged or unusual bleeding, bruising, purpura, dark stool) should be explained

to the patients and they should be advised to stop medication and consult physician immediately if the symptoms occur. It should be used with caution in patients with renal or hepatic impairment. **Usage in pregnancy:** FDA Pregnancy Category B: No adequate and well controlled studies in pregnant women. Therefore risk-benefits should be considered. **Usage in lactation:** Studies in rats have shown ticlopidine is excreted in milk. However, it is not known whether this drug is excreted in human milk. **Usage in pediatrics:** Safety and efficacy have not been established. **Usage in elderly:** Clearance of ticlopidine is lower in elderly patients. However, no overall differences in efficacy or safety were observed between elderly patients and younger adults.

Drug interactions:

Non-steroidal anti inflammatory drugs (NSAIDs), antiplatelet drugs, oral anticoagulants, heparin and salicylic derivatives - may increase haemorrhagic risk.
Theophylline - concomitant administration resulted in significant increase in the theophylline elimination half-life and a reduction in total plasma clearance of theophylline.
Digoxin - ticlopidine decreases plasma digoxin levels (15%).
Antacid - decreases plasma levels of ticlopidine by 18%. Both drugs should be administered at least 1-2 hours apart.
Cimetidine - decreases clearance of ticlopidine by 50%.
Phenytoin - several cases of elevated phenytoin level with associated somnolence and lethargy have been reported following co-administration. Phenytoin blood levels should be remeasured.
Propranolol - Although in vitro studies demonstrated that ticlopidine does not alter the plasma protein binding of propranolol, in vivo studies have not been studied. Therefore, caution should be exercised in coadministering this drug with ticlopidine.

Warning

Haematological and haemorrhagic adverse effects with commonly severe and sometimes fatal consequences have been observed (see adverse effects). These severe effects are not uncommonly associated with:

- inadequate monitoring, delayed diagnosis and inappropriate therapeutic measures for adverse effects;
- concomitant administration of anticoagulants or antiplatelet agents such as aspirin and nonsteroidal antiinflammatory drugs.

Contraindications:

It is contraindicated in patients hypersensitive to ticlopidine, history of leucopenia, thrombocytopenia or agranulocytosis, hemopathies & haemorrhagic diathesis and other blood diseases involving prolonged bleeding time, organic lesions (active gastroduodenal ulcer or in the acute phase of haemorrhagic cerebrovascular accident). It should not be given to patients receiving aspirin, nonsteroidal antiinflammatory agents, antiplatelets or anticoagulants.

Dosage and Administration:

Adults: 1-2 tablets daily, with meals.

Treatment of overdosage:

Overdosage may cause severe gastrointestinal intolerance and increased bleeding time. Recommended treatment consists of monitoring the patients and instituting supportive measures, as needed. For bleeding complications, platelet transfusion may be helpful and other measures to control bleeding must be employed.

Storage	: Keep container tightly closed. Store in a cool, dry place below 30°C Protect from light.
Presentation	: 3 X 10's/ blister
Shelflife	: 3 years
Product registration holder & Manufacturer	: PRIME PHARMACEUTICAL SDN. BHD. 1505, Lorong Perusahaan Utama 1, Taman Perindustrian Bukit Tengah, 14000 Bukit Mertajam, Penang, Malaysia.

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